Prathyusha Sadhu

Senior Frontend Engineer | Angular Lead | Frontend Architect

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PROFILE SUMMARY

Senior Frontend Engineer with 9+ years building enterprise-grade Angular applications, including 3 years leading a frontend team in Canada. Specializes in TypeScript, RxJS, and NgRx, with recent focus on integrating Generative AI, conversational interfaces, and voice agents into production systems. Experienced working directly with North American clients and cross-border teams, consistently delivering measurable performance gains — Lighthouse scores of 90+, LCP improvements from 4s to under 2.2s. Known for turning ambiguous requirements into clean, scalable frontend solutions.

SKILLS

Frameworks & Libraries: Angular (v8–19), TypeScript, JavaScript, jQuery, Bootstrap, Material UI (MUI), Tailwind CSS, NgRx, RxJS

Web Technologies: HTML5, CSS3/SASS, SCSS, AJAX, XML, JSON, RESTful Web Services, SOAP

AI & Prompt Engineering: Conversational AI, Voice Agents, Prompt Engineering, Claude, GitHub Copilot, Cursor, Codex

Cloud & DevOps: Azure, AEM, Docker, CI/CD (GitHub Actions, Azure DevOps, Jenkins), Webpack, NPM, Grunt, Maven

Testing Tools: Jasmine, Karma, Jest, Cypress, React Testing Library (RTL), Postman, SoapUI

Version Control & Tracking: GIT, Bitbucket, Confluence, JIRA, ServiceNow

Design & Collaboration: Figma, Responsive Design, Web Accessibility

IDEs: Visual Studio Code, IntelliJ IDEA, Eclipse IDE, Notepad++, EditPlus

CERTIFICATIONS

- Generative AI — Outskill
- Understanding Agentic AI — Agent Academy

WORK EXPERIENCE

Technology Analyst — Infosys Limited

Sep 2021 – Mar 2026

Client: C Spire Wireless (Telecom) • Hyderabad, India (Sep 2021 – Apr 2023) → Canada (May 2023 – Mar 2026)

- **Developed and enhanced** enterprise Angular applications for a telecom client, building reusable UI components and integrating REST APIs using TypeScript, RxJS and NgRx.
- Designed and implemented **30+ reusable UI components** to reduce code duplication and accelerate delivery.
- Implemented **dynamic image resizing and optimization** for AEM DAM background/hero images using Dynamic Media with configurable width and quality parameters, reducing image payload and improving page performance and Core Web Vitals.
- Investigated and resolved **Google Pay and Apple Pay token integration issues** across environments using Splunk, browser debugging, and API/network analysis.
- Implemented SEO requirements including page titles, meta descriptions, canonical URLs, and **Google Tag Manager** integration.
- Translated **Figma designs** into responsive Angular components across desktop, tablet, and mobile breakpoints.
- Implemented **NgRx-based** state management for complex business workflows, lowering state-related defects and improving maintainability.
- Used **SonarQube** to maintain code quality against a required **90% code coverage quality gate**, addressing duplicate code and coverage gaps to keep builds passing.
- Used **Postman and SoapUI** for REST API testing and debugging.
- **Collaborated** with product owners, architects, and stakeholders to streamline requirements and delivery.
- Executed unit and integration testing using Jasmine, Karma and Jest, keeping components above the **90% coverage threshold** required by SonarQube.
- Established and maintained **CI/CD pipelines** (GitHub Actions / Azure DevOps) for automated build, testing, and deployment.

- Achieved **Lighthouse performance scores of 90+**, improved LCP from ~4s to under 2.2s, and reduced bundle size by 30% through lazy loading, code-splitting and module optimization.

Senior Software Engineer — Mindtree Limited **Jul 2016 – Sep 2021**

Client engagements: Microsoft (Jun 2018 – Sep 2021) • Capillary (Mar 2018 – Jun 2018) • Toyota (Mar 2017 – Feb 2018) • Gemalto (Nov 2016 – Mar 2017)

- Developed product front-end using **HTML5, CSS3, Bootstrap**, and jQuery. Created responsive user interfaces for optimal cross-device experience.
- Led frontend modernization initiatives for enterprise platforms, developing modular, reusable Angular components.
- Built **component libraries** adopted across multiple business modules to accelerate delivery and maintain consistency.
- Improved **SEO** and page speed, achieving an 18% increase in organic traffic and raising page speed scores from 55 to 88.
- Applied **Agile/Scrum** principles to project execution, actively participating in sprint planning, daily stand-ups, and sprint reviews.
- Communicated directly with clients to understand their needs, provide timely project updates, and maintain strong client relationships.

PROJECTS

Jarvis — Intelligent AI Personal Assistant (In Development)

- Designing an AI-powered personal assistant integrating voice-controlled task management, calendar and email automation, and intelligent reminders using speech recognition and NLP.
- Implementing context-aware conversational AI with memory management and agentic AI workflows to autonomously execute multi-step tasks.
- Using Angular, TypeScript, Web Speech API and Claude AI API to enable real-time synchronization and multi-modal interactions.

q-live — Multi-Tenant AI Voice Agent for Clinic Appointment Booking

- Designed a multi-tenant voice AI architecture handling phone-based clinic appointment booking, using SIP telephony, LiveKit, Sarvam AI for Telugu speech-to-text/text-to-speech, and Groq LLM for conversation and appointment logic.
- Built on Supabase to support multiple clinics per deployment, each with its own phone number, greeting, prompt configuration, and appointment data.
- Explored patent filing for the architecture, including India-specific patentability requirements under Section 3(k).

Stock Analyzer — Indian Stock Market Dashboard

- Built an independent Python/Streamlit application for market-wide scanning of Indian equities, with sector-grouped results and a rule-based buy/sell scoring engine.
- Implemented candlestick pattern detection, EMA/RSI overlays, and swing-trading analysis across the Nifty Total Market (~750 stocks) and a combined NSE+BSE universe (~7,800 stocks).
- Deployed on Render; repo: https://github.com/prathyushasadhu/stock-analyzer.

Multi-Model AI Chat & Customer Support Platform

- Built an intelligent chatbot handling **10,000+ monthly conversations** with dynamic model switching between Claude, GPT-4 and domain models to improve response relevance.
- Implemented prompt engineering and role-based system prompts achieving **85% first-contact resolution rate** and integrated sentiment analysis to reduce escalation time by 45%.

AI-Assisted Performance Optimization Engine

- Built an automated AI tool to identify frontend performance bottlenecks and recommend code-splitting and lazy-loading strategies.
- Implemented an ML-based bundle analyzer and achieved an average **42% improvement in Core Web Vitals** across multiple enterprise applications.

EDUCATION

B.Tech / B.E. — Computer Science and Engineering (CSE)

Koneru Lakshmaiah Education Foundation, Guntur • 2012 – 2016 • Grade: 8.9/10